

VERILOG מעבדה 4

מגישים:

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1)

Permutation:

```
1 // Initial Permutation
2 module Permutation
3 (
4     input reg [63:0] Data_In,
5     output [63:0] Y
6 );
7     assign Y[0] = Data_In[57] ;
8     assign Y[1] = Data_In[49] ;
9     assign Y[2] = Data_In[41] ;
10    assign Y[3] = Data_In[33] ;
11    assign Y[4] = Data_In[25] ;
12    assign Y[5] = Data_In[17] ;
13    assign Y[6] = Data_In[9] ;
14    assign Y[7] = Data_In[1] ;
15    assign Y[8] = Data_In[59] ;
16    assign Y[9] = Data_In[51] ;
17    assign Y[10] = Data_In[43] ;
18    assign Y[11] = Data_In[35] ;
19    assign Y[12] = Data_In[27] ;
20    assign Y[13] = Data_In[19] ;
21    assign Y[14] = Data_In[11] ;
22    assign Y[15] = Data_In[3] ;
23    assign Y[16] = Data_In[61] ;
24    assign Y[17] = Data_In[53] ;
25    assign Y[18] = Data_In[45] ;
26    assign Y[19] = Data_In[37] ;
27    assign Y[20] = Data_In[29] ;
28    assign Y[21] = Data_In[21] ;
29    assign Y[22] = Data_In[13] ;
30    assign Y[23] = Data_In[5] ;
31    assign Y[24] = Data_In[63] ;
32    assign Y[25] = Data_In[55] ;
33    assign Y[26] = Data_In[47] ;
34    assign Y[27] = Data_In[39] ;
35    assign Y[28] = Data_In[31] ;
36    assign Y[29] = Data_In[23] ;
37    assign Y[30] = Data_In[15] ;
38    assign Y[31] = Data_In[7] ;
39    assign Y[32] = Data_In[56] ;
40    assign Y[33] = Data_In[48] ;
41    assign Y[34] = Data_In[40] ;
42    assign Y[35] = Data_In[32] ;
43    assign Y[36] = Data_In[24] ;
44    assign Y[37] = Data_In[16] ;
45    assign Y[38] = Data_In[8] ;
46    assign Y[39] = Data_In[0] ;
47    assign Y[40] = Data_In[58] ;
48    assign Y[41] = Data_In[50] ;
49    assign Y[42] = Data_In[42] ;
50    assign Y[43] = Data_In[34] ;
51    assign Y[44] = Data_In[26] ;
52    assign Y[45] = Data_In[18] ;
53    assign Y[46] = Data_In[10] ;
54    assign Y[47] = Data_In[2] ;
55    assign Y[48] = Data_In[60] ;
56    assign Y[49] = Data_In[52] ;
57    assign Y[50] = Data_In[44] ;
58    assign Y[51] = Data_In[36] ;
59    assign Y[52] = Data_In[28] ;
60    assign Y[53] = Data_In[20] ;
61    assign Y[54] = Data_In[12] ;
62    assign Y[55] = Data_In[4] ;
63    assign Y[56] = Data_In[62] ;
64    assign Y[57] = Data_In[54] ;
65    assign Y[58] = Data_In[46] ;
66    assign Y[59] = Data_In[38] ;
67    assign Y[60] = Data_In[30] ;
68    assign Y[61] = Data_In[22] ;
69    assign Y[62] = Data_In[14] ;
70    assign Y[63] = Data_In[6] ;
71
72 endmodule
```

Inverse Permutation:

```
1  // Inverse Permutation
2  module In_Permutation
3  (
4      input    [63:0] Y,
5      output   [63:0] Data_Out
6  );
7
8      assign Data_Out[0]  = Y[39] ;
9      assign Data_Out[1]  = Y[7]   ;
10     assign Data_Out[2]  = Y[47]  ;
11     assign Data_Out[3]  = Y[15]  ;
12     assign Data_Out[4]  = Y[55]  ;
13     assign Data_Out[5]  = Y[23]  ;
14     assign Data_Out[6]  = Y[63]  ;
15     assign Data_Out[7]  = Y[31]  ;
16     assign Data_Out[8]  = Y[38]  ;
17     assign Data_Out[9]  = Y[6]   ;
18     assign Data_Out[10] = Y[46]  ;
19     assign Data_Out[11] = Y[14]  ;
20     assign Data_Out[12] = Y[54]  ;
21     assign Data_Out[13] = Y[22]  ;
22     assign Data_Out[14] = Y[62]  ;
23     assign Data_Out[15] = Y[30]  ;
24     assign Data_Out[16] = Y[37]  ;
25     assign Data_Out[17] = Y[5]   ;
26     assign Data_Out[18] = Y[45]  ;
27     assign Data_Out[19] = Y[13]  ;
28     assign Data_Out[20] = Y[53]  ;
29     assign Data_Out[21] = Y[21]  ;
30     assign Data_Out[22] = Y[61]  ;
31     assign Data_Out[23] = Y[29]  ;
32     assign Data_Out[24] = Y[36]  ;
33     assign Data_Out[25] = Y[4]   ;
34     assign Data_Out[26] = Y[44]  ;
35     assign Data_Out[27] = Y[12]  ;
36     assign Data_Out[28] = Y[52]  ;
37     assign Data_Out[29] = Y[20]  ;
```

```
37     assign Data_Out[29] = Y[20] ;
38     assign Data_Out[30] = Y[60] ;
39     assign Data_Out[31] = Y[28] ;
40     assign Data_Out[32] = Y[35] ;
41     assign Data_Out[33] = Y[3]  ;
42     assign Data_Out[34] = Y[43] ;
43     assign Data_Out[35] = Y[11] ;
44     assign Data_Out[36] = Y[51] ;
45     assign Data_Out[37] = Y[19] ;
46     assign Data_Out[38] = Y[59] ;
47     assign Data_Out[39] = Y[27] ;
48     assign Data_Out[40] = Y[34] ;
49     assign Data_Out[41] = Y[2]  ;
50     assign Data_Out[42] = Y[42] ;
51     assign Data_Out[43] = Y[10] ;
52     assign Data_Out[44] = Y[50] ;
53     assign Data_Out[45] = Y[18] ;
54     assign Data_Out[46] = Y[58] ;
55     assign Data_Out[47] = Y[26] ;
56     assign Data_Out[48] = Y[33] ;
57     assign Data_Out[49] = Y[1]  ;
58     assign Data_Out[50] = Y[41] ;
59     assign Data_Out[51] = Y[9]  ;
60     assign Data_Out[52] = Y[49] ;
61     assign Data_Out[53] = Y[17] ;
62     assign Data_Out[54] = Y[57] ;
63     assign Data_Out[55] = Y[25] ;
64     assign Data_Out[56] = Y[32] ;
65     assign Data_Out[57] = Y[0]  ;
66     assign Data_Out[58] = Y[40] ;
67     assign Data_Out[59] = Y[8]  ;
68     assign Data_Out[60] = Y[48] ;
69     assign Data_Out[61] = Y[16] ;
70     assign Data_Out[62] = Y[56] ;
71     assign Data_Out[63] = Y[24] ;
72
73     endmodule
```

Testbench Permutation:

```
2  `timescale 1ns/1ps
3  module TB_Permutation();
4
5      reg [63:0] tb_Data_In;
6      wire [63:0] tb_Y;
7      wire [63:0] tb_Data_Out;
8
9      initial begin
10
11          $display ("time\t\t data in\t\t\t y\t\t data out");
12          $monitor ("%8g\t\t %h    %h    %h",
13                  $time, tb_Data_In, tb_Y, tb_Data_Out);
14          tb_Data_In = 64'h0123456789abcdef;
15          #50 tb_Data_In = 64'h123456789abcdef0;
16          #50 tb_Data_In = 64'h23456789abcdef01;
17          #50 tb_Data_In = 64'h3456789abcdef012;
18          #50 tb_Data_In = 64'h456789abcdef0123;
19          #50 tb_Data_In = 64'h56789abcdef01234;
20          #50 tb_Data_In = 64'h6789abcdef012345;
21          #50 tb_Data_In = 64'h789abcdef0123456;
22          #50 tb_Data_In = 64'h89abcdef01234567;
23          #50 tb_Data_In = 64'h9abcdef012345678;
24          #50 tb_Data_In = 64'habcdef0123456789;
25          #50 tb_Data_In = 64'hbcdef0123456789a;
26          #50 tb_Data_In = 64'hcdef0123456789ab;
27          #50 tb_Data_In = 64'hdef0123456789abc;
28          #50 tb_Data_In = 64'hef0123456789abcd;
29          #50 tb_Data_In = 64'hf0123456789abcde;
30          #50 $finish;
31      end
32
33      Permutation DUT1
34      (
35          .Data_In    (tb_Data_In    ),
36          .Y          (tb_Y )
37      );
38
39      In_Permutation DUT2
40      (
41          .Y          (tb_Y ),
42          .Data_Out   (tb_Data_Out )
43      );
44
45      endmodule
```

log of the \$monitor tracking:

```
# time          data in          y          data out
#      0      0123456789abcdef  cc00ccfff0aaf0aa  0123456789abcdef
#     50     123456789abcdef0  ccff6600f0aa7855  123456789abcdef0
#    100     23456789abcdef01  660066ff78557855  23456789abcdef01
#    150     3456789abcdef012  66ff330078553caa  3456789abcdef012
#    200     456789abcdef0123  330033ff3caa3caa  456789abcdef0123
#    250     56789abcdef01234  33ff99003caa1e55  56789abcdef01234
#    300     6789abcdef012345  990099ffe551e55  6789abcdef012345
#    350     789abcdef0123456  99ffcc001e550faa  789abcdef0123456
#    400     89abcdef01234567  cc00ccff0faa0faa  89abcdef01234567
#    450     9abcdef012345678  ccff66000faa8755  9abcdef012345678
#    500     abcdef0123456789  660066ff87558755  abcdef0123456789
#    550     bcdef0123456789a  66ff33008755c3aa  bcdef0123456789a
#    600     cdef0123456789ab  330033ffc3aac3aa  cdef0123456789ab
#    650     def0123456789abc  33ff9900c3aae155  def0123456789abc
#    700     ef0123456789abcd  990099ffe155e155  ef0123456789abcd
#    750     f0123456789abcde  99ffcc00e155f0aa  f0123456789abcde
# ** Note: $finish      : Z:/verilog/lab_4/TB_Permutation.v(31)
#   Time: 800 ns  Iteration: 0  Instance: /TB_Permutation
```

waveforms:

